

Problem 1: Review of Macroscopic Balances and Numerical Methods Used in Commercial Process Simulators.

Colebrook Equation:

$$F = \frac{1}{\sqrt{f}} + 1.737 \ln \left[\frac{1.256}{Re\sqrt{f}} + \frac{k}{3.707D} \right] = 0$$

1. Solve by the Newton method to obtain the friction factor.

To solve this problem, we must find the first derivative of the Colebrook Equation:

$$\frac{dF}{df} = -\frac{1}{2f^{\frac{3}{2}}} - 1.737 \left[\frac{1}{\frac{1.256}{Re\sqrt{f}} + \frac{k}{3.707D}} \right] \left(\frac{1.256}{2Re \cdot f^{\frac{3}{2}}} \right)$$

Using the formula:

$$x_{k+1} = x_k - \left(\frac{F(x_k)}{F'(x_k)} \right)$$

Where $x = f$, and k is the number of iterations, we can solve for f using Excel for calculations. As shown below, f was calculated to be 0.004201.

k	Xk	f(Xk)	f'(Xk)	Xk+1	Xk+1-Xk	ξ
0	0.002	7.245539	-5608.85	0.003291804	0.001291804	0.645902099
1	0.003292	2.097879	-2661.61	0.004080003	0.000788199	0.239442851
2	0.00408	0.239114	-1931.16	0.004203822	0.000123819	0.030347777
3	0.004204	-0.00461	-1846.81	0.004201328	-2.49375E-06	0.000593209
4	0.004201	0.000197	-1848.45	0.004201435	1.06722E-07	2.5402E-05
5	0.004201	-8.4E-06	-1848.38	0.004201431	-4.51993E-09	1.07581E-06
6	0.004201	3.54E-07	-1848.38	0.004201431	1.91515E-10	4.55834E-08
					f =	0.004201431

Our answer was obtained in 6 iterations and over the tolerance, ξ , of 1E-06.

2. Repeat by using the Secant Method.

Unlike the Newton's method, no derivative is required here. Thus, we can move forward with this method by using the following formula:

$$x_{k+1} = x_k - F(x_k) \left[\frac{x_k - x_{k-1}}{F(x_k) - F(x_{k-1})} \right]$$

Where two initial guesses are needed. Again, using Excel, the following computations and iterations were made to obtain f , as shown below.

k	Xk	Xk-1	f(Xk)	f(Xk-1)	Xk-Xk-1	f(Xk)-f(Xk-1)	Xk+1	ξ
0	0.002	0.001	7.245539	16.85305558	0.001	-9.607516142	0.002754	0.5
1	0.002754	0.002	3.797675	7.24553944	0.000754153	-3.447864421	0.003585	0.273824
2	0.003585	0.002754	1.336034	3.797675019	0.000830668	-2.461640754	0.004036	0.231718
3	0.004036	0.003585	0.329098	1.336034264	0.000450838	-1.006936324	0.004183	0.111714
4	0.004183	0.004036	0.035615	0.32909794	0.000147348	-0.293482554	0.004201	0.035225
5	0.004201	0.004183	0.001047	0.035615385	1.78813E-05	-0.034568073	0.004201	0.004257
6	0.004201	0.004201	3.42E-06	0.001047312	5.41751E-07	-0.00104389	0.004201	0.000129
7	0.004201	0.004201	3.3E-10	3.42191E-06	1.77588E-09	-3.42158E-06	0.004201	4.23E-07
f =							0.004201431	

The same value of f , 0.004201, was obtained at the same tolerance.

3. Repeat by using the Direct Substitution Method.

For this method, we needed to put f in terms ψ and of a new function, $g(\psi)$, to satisfy:

$$\varphi = g(\varphi)$$

Here, we can define:

$$\varphi = \frac{1}{\sqrt{f}} \Rightarrow f = \frac{1}{\varphi^2}$$

Which transforms our equation to a tear equation:

$$\varphi = -1.737 \ln \left[\frac{1.256}{Re} \varphi + \frac{k}{3.707D} \right] = g(\varphi)$$

Thus, the above condition is satisfied. Now, we can solve for ψ , then f , using Excel.

k	Xk	g(Xk)	ξ		
0	10	15.71891	0.571891		
1	15.71891	15.41338	0.019437		
2	15.41338	15.42842	0.000975		
3	15.42842	15.42767	4.81E-05		
4	15.42767	15.42771	2.38E-06		
5	15.42771	15.42771	1.17E-07	f = 1/ψ^2 =	0.004201431

4. Repeat by using the Broyden Method.

Since this system is only in 1-dimension, we can use the following formula:

$$\varphi_{k+1} = \varphi_k - \mathbf{B}(\varphi_k)\mathbf{F}(\varphi_k)$$

Where:

$$\mathbf{B}(\varphi_k) = \mathbf{J}(\varphi_k)^{-1} = \left(\frac{dg}{d\varphi}\right)^{-1} \text{ and } \mathbf{F}(\varphi_k) = \varphi - g(\varphi) = 0$$

Because the original Colebrook Equation was set to zero, we used that equation, along with the first derivative, in our Excel computations, shown below:

k	Xk	F(Xk)	F'(Xk)	B(Xk)	Xk+1	ξ
0	0.003	2.964146	-3057.87	-0.00032702	0.003969349	-0.323116311
1	0.003969	0.466435	-2012.14	-0.00049698	0.004201159	-0.058399974
2	0.004201	0.000524	-1848.56	-0.00054096	0.004201442	-6.74933E-05
3	0.004201	-2.2E-05	-1848.37	-0.00054102	0.00420143	2.85627E-06
4	0.004201	9.4E-07	-1848.38	-0.00054101	0.004201431	-1.21028E-07
					f =	0.004201431

We obtain $f = 0.004201$ within our set tolerance.

5. Repeat by using the Wegstein Acceleration Method.

For this method, we used the tear equation derived in part 3. Then, the calculations were performed using the derived Wegstein method that gives the following:

$$x_{k+1} = \omega \cdot g(x_k) + (1 - \omega)x_k$$

Where ω is the acceleration factor, in which

$$\omega = \frac{1}{[1 - s]} \text{ and } s = \frac{g(x_k) - g(x_{k-1})}{x_k - x_{k-1}}$$

Using these equations in Excel for our computations, we obtained:

k	Xk-1	Xk	g(Xk-1)	g(Xk)	s	ω	ξ
0	15	16	15.44896	15.39966636	-0.049297113	0	0.025953
1	16	15.39967	15.39967	15.42909292	-0.049017007	0	0.037002
2	0.0035	0.004	16.43057	16.4305292	-0.087987484	0.91912822	0.999787
3	0.004	15.10209	16.43053	15.4438663	-0.065350195	0.938658485	0.999741
4	15.10209	15.4229	15.44387	15.42794471	-0.049628652	0.952717895	0.021121
5	15.4229	15.42771	15.42794	15.42770734	-0.049398519	0.952926826	0.000312
6	15.42771	15.42771	15.42771	15.42770728	-0.049395144	0.952929891	6.83E-08
					f = 1/ψ^2 =	0.004201431	

The f value is the same as in previous methods and within tolerance.

6. What is the rate of heat transfer to the air?

For this problem we must start with an energy balance assuming steady state and no potential energy:

$$\frac{dU}{dt} = \Delta \left(H + \frac{1}{2} v^2 \right) = Q + W_s$$

Since there is no work done on or by the pipe, we can eliminate W and distribute Δ :

$$\Delta H + \frac{1}{2} (v_2^2 - v_1^2) = Q$$

Substituting for $C_p \Delta T$ for ΔH , we can come to the equation:

$$C_p \Delta T + \frac{1}{2} v_2^2 - \frac{1}{2} v_1^2 = Q$$

However, to find each v , we must first find the pressure drop, ΔP , densities, ρ_i , and the mass velocity, G . We can find ΔP in terms of a ratio, x , shown in the equation below:

$$-G^2 \ln \left(x \frac{T_1}{T_2} \right) + \frac{M_w P_1^2}{2RT_a} (x^2 - 1) + G^2 \left(\frac{2Lf}{D} \right) = 0$$

Where: $G = \rho v = \frac{Re \cdot \mu}{D}$, $x = \frac{P_2}{P_1}$, $T_a = \frac{(T_1 + T_2)}{2}$, L = length of the pipe, f = friction factor, D = diameter of the pipe, M_w = molecular weight, and R = the universal gas constant.

To solve for x , we used GOALSEEK in Excel. We found this value to be:

$$x = 0.61007$$

Using $P_2 = x \cdot P_1$, we found ΔP to be:

$$\Delta P = P_2 - P_1 = (6.47 - 20) \text{ psig} = -13.53 \text{ psig}$$

Next, we used the ideal gas law to find ρ_1 and ρ_2 by:

$$\rho_i = \frac{P_i}{RT_i}, \text{ where } \rho_1 = 2.832 \frac{\text{kg}}{\text{m}^3} \text{ and } \rho_2 = 1.453 \frac{\text{kg}}{\text{m}^3}$$

Now, using the relation $G = \rho v$, we can find v_1 and v_2 by:

$$v_i = G \rho_i, \text{ where } v_1 = 76.61 \frac{\text{m}}{\text{s}} \text{ and } v_2 = 149.29 \frac{\text{m}}{\text{s}}$$

All constants, listed in the spreadsheet below, were found using Engineering Toolbox.

Since our Q equation is in terms of energy per mass, we must multiply it by the mass flow rate, $\dot{m}$. To find this, we can use G and the area of the pipe.

$$\dot{m} = G \frac{\pi D^2}{4} = 0.121 \frac{\text{kg}}{\text{s}}$$

Finally, using the energy balance and $\dot{m}$, we obtain a value for $\dot{Q}$ as:

$$\dot{Q} = 0.121 \frac{\text{kg}}{\text{s}} \left\{ 1.007 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} (55.55 \text{K}) + \left(\frac{1}{2} (149.29)^2 - \frac{1}{2} (76.61)^2 \right) \frac{\text{m}^2}{\text{s}^2} \left(\frac{\text{J} \cdot \text{s}^2}{\text{kg} \cdot \text{m}^2} \right) \left(\frac{\text{kJ}}{1000 \text{J}} \right) \right\}$$

$$\dot{Q} = 7.77 \frac{\text{kJ}}{\text{s}} = 26511.8 \frac{\text{Btu}}{\text{hr}}$$

Spreadsheet for Calculations of G and Q:

Computation for Q				
$\dot{Q} = G \frac{\pi D^2}{4} [C_p \Delta T + \frac{1}{2} \Delta v^2]$				
Quantity	Value	Units		
Re	318000	--		
D	0.027	m		
L	10.973	m		
f	0.004201	--		
μ	1.82E-05	Pa*s		
G	216.976	Pa*s/m		
m (mass flow rate)	0.12111303	kg/s		
T1	294.261	K		
T2	349.817	K		
ΔT	55.556	K		
P1	20	psig		
P2	6.47114605	psig		
ΔP	13.5288539	psig		
R	8314.462	m ³ *Pa/K*kmol		
ρ_1	2.832	kg/m ³		
ρ_2	1.453	kg/m ³		
v_1	76.615	m/s		
v_2	149.292	m/s		
$C_p(T_a)$	1.007	kJ/kg*K		
x (P2/P1)	0.610	--		
Q	7.7698395	kJ/s	26511.7926	Btu/hr
$-G^2 \ln \left(x \frac{T_1}{T_2} \right) + \frac{M_w P_1^2}{2RT_a} (x^2 - 1) + G^2 \left(\frac{2Lf}{D} \right) = 0$				
Calculations for x				
Ta	322.039	K		
Mw	28.9647	kg/kmol		
x(guess)	0.61007			
GOALSEEK	0.00			